

The Evolving Landscape of LLM- and VLM-Integrated Reinforcement Learning

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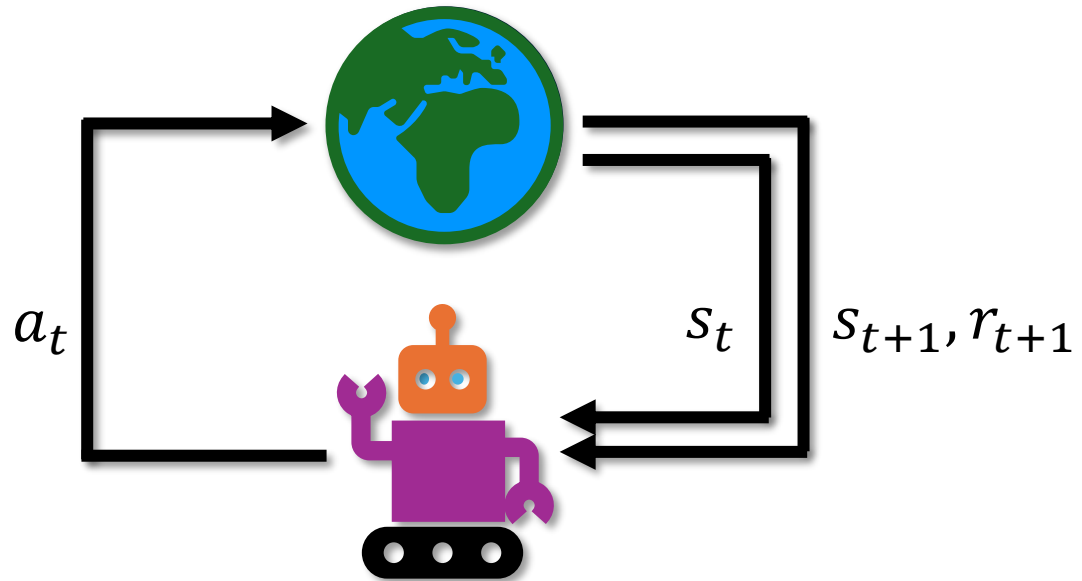


**The Intelligent
Robot Learning
Laboratory**

Preliminaries

Reinforcement Learning & Foundation Models

Reinforcement Learning (RL)



- Sequential decision-making
- Policy $\pi(a|s)$
- Maximize expected discounted return

$$E_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \right]$$

Reinforcement Learning (RL)

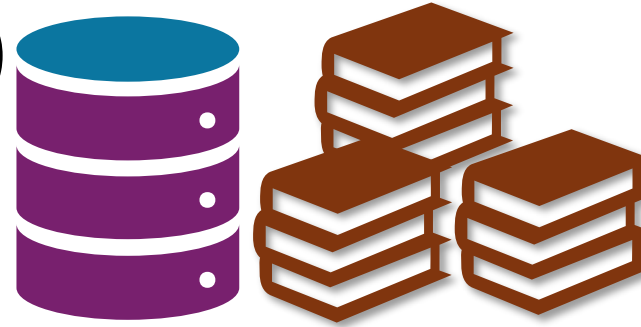
Main Strengths of RL

- Requires only a scalar reward signal to learn complex behaviours
- Learns through interaction (trial-and-error)
- Transfer learned policies to related tasks
- Adapts to non-stationary environments

Main Weaknesses of RL

- Sample inefficient
- Credit assignment in long-horizon tasks with sparse rewards is difficult.
- Reward design pitfalls
 - May be difficult to specify (e.g., kneeling task)
 - May lead to unintended/unsafe behaviour

Foundation Models (FMs)

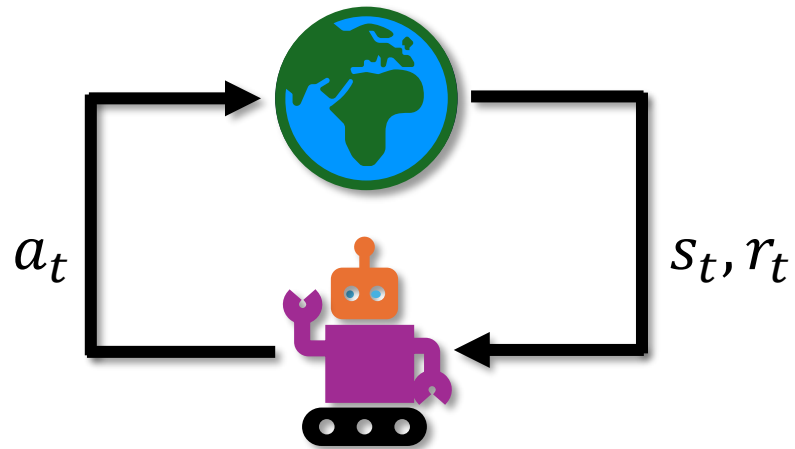


Human Learning

- Humans learn through trial and error, but...
- Humans also rely on external sources of information (parents, teachers, books, etc.) to shape their knowledge.



Can Large Language Models (LLMs) and Vision-Language Models (VLMs) be integrated into RL?



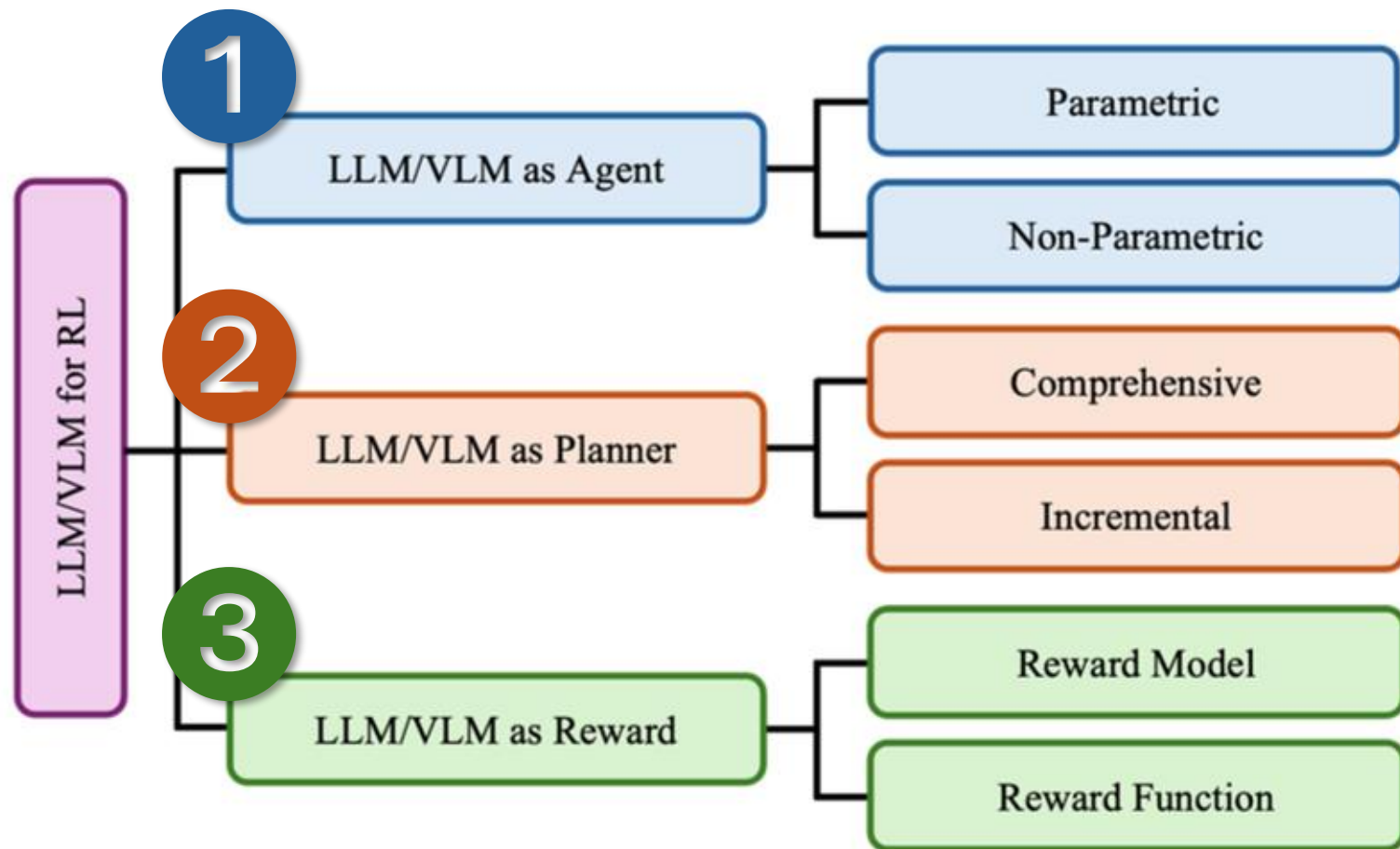
Capabilities

- Natural-language interface
- Rich priors
- Emergent
 - Strong generalization
 - Chain-of-thought reasoning*
 - In-context learning (gradient-free)
 - Self-critique and self-refinement

Our Survey

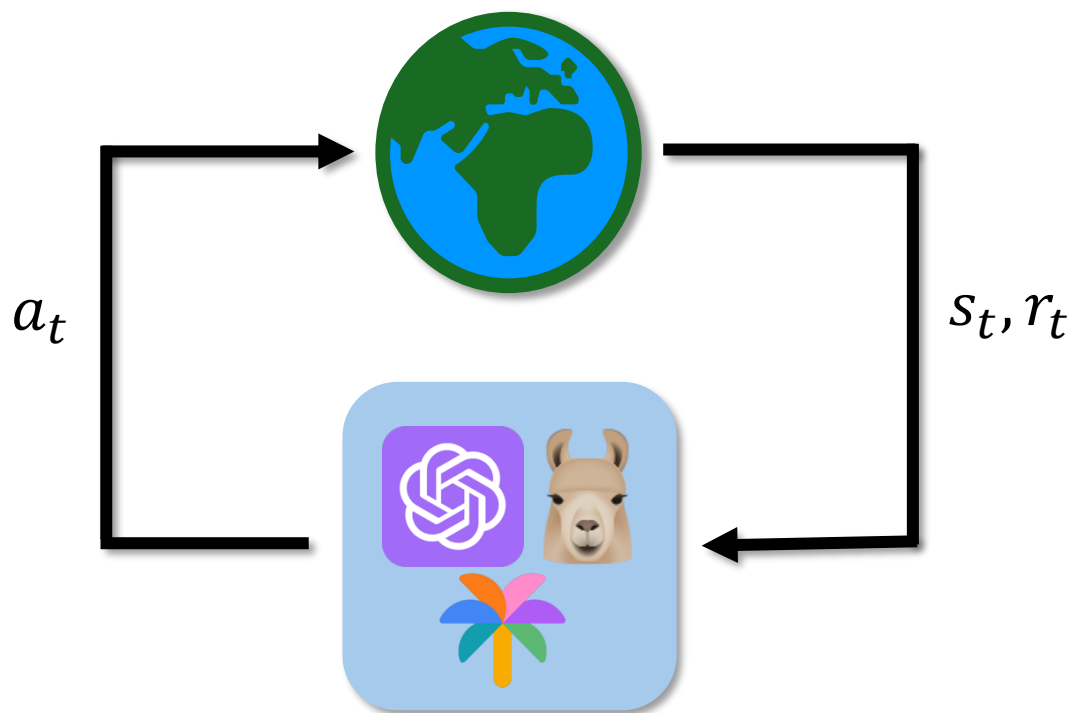
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Taxonomy



1

LLM/VLM as Agent



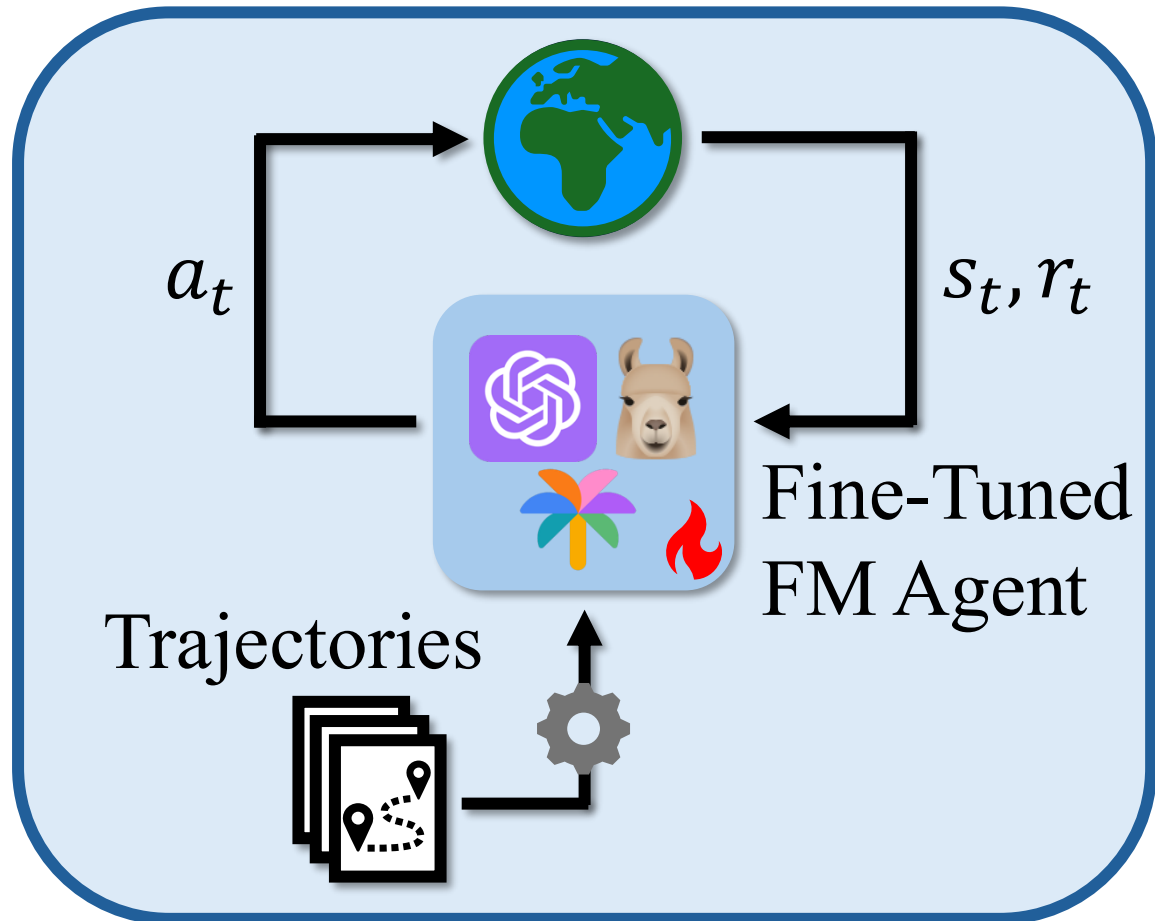
(Sub-)Trajectories



$s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, r_3, \dots$

1

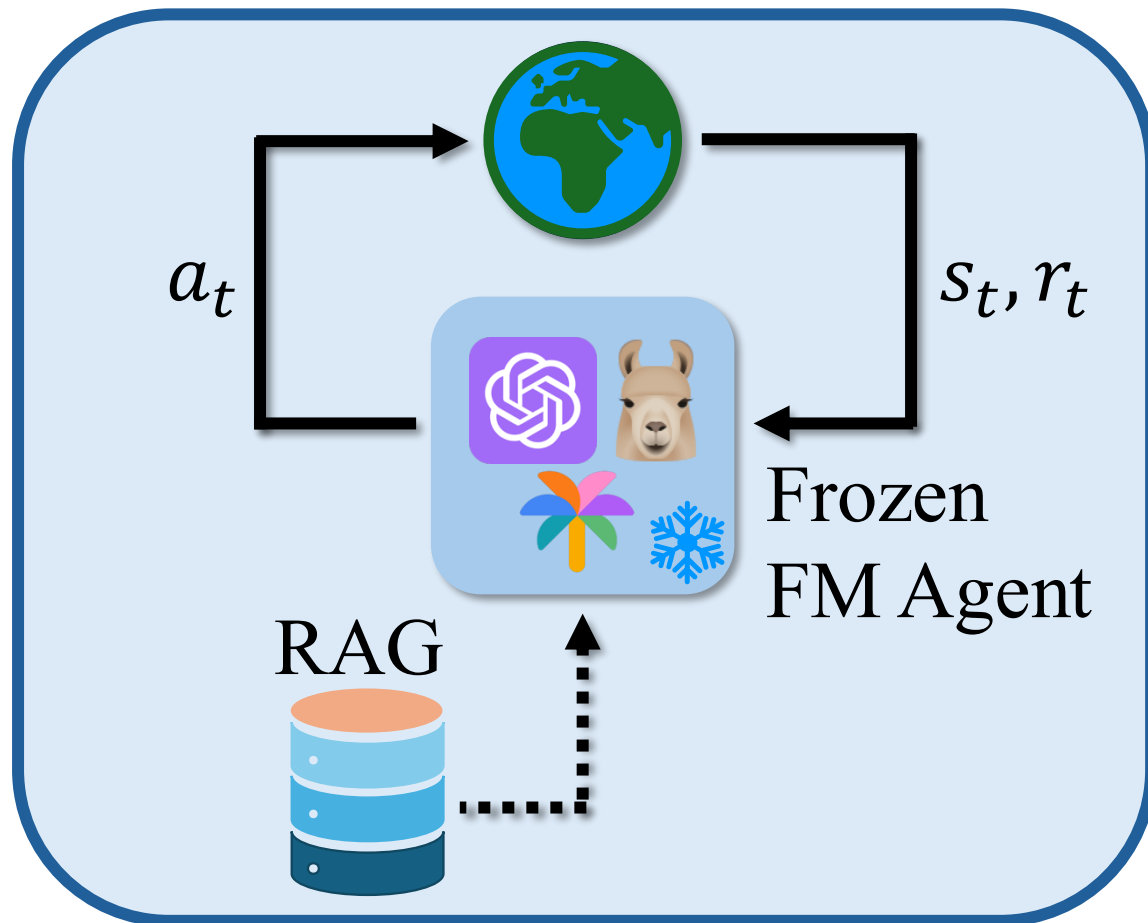
LLM/VLM as Agent: Parametric



- Parameter efficient fine-tuning techniques
 - LoRA
 - Adapters
 - Prefix tuning

1

LLM/VLM as Agent: Non-Parametric



2

LLM/VLM as Planner

Hey LLM! Get me a can of soda, *please*? 😊

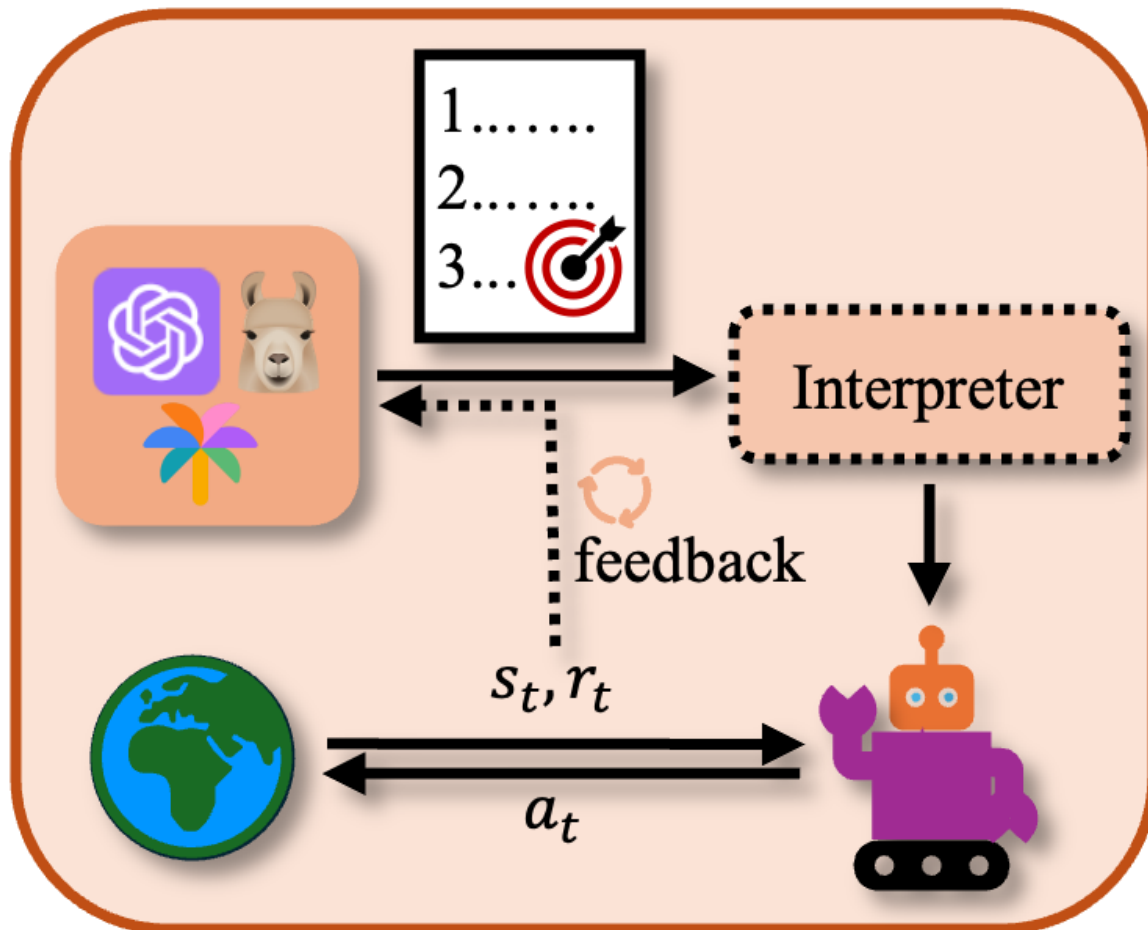


Sure thing! A plan to “get you a can of soda” is:

1. Locate soda can.
2. Navigate to <LOCATION>.
3. Grasp soda can.
4. Navigate to <PERSON>.
5. Place soda can.

2

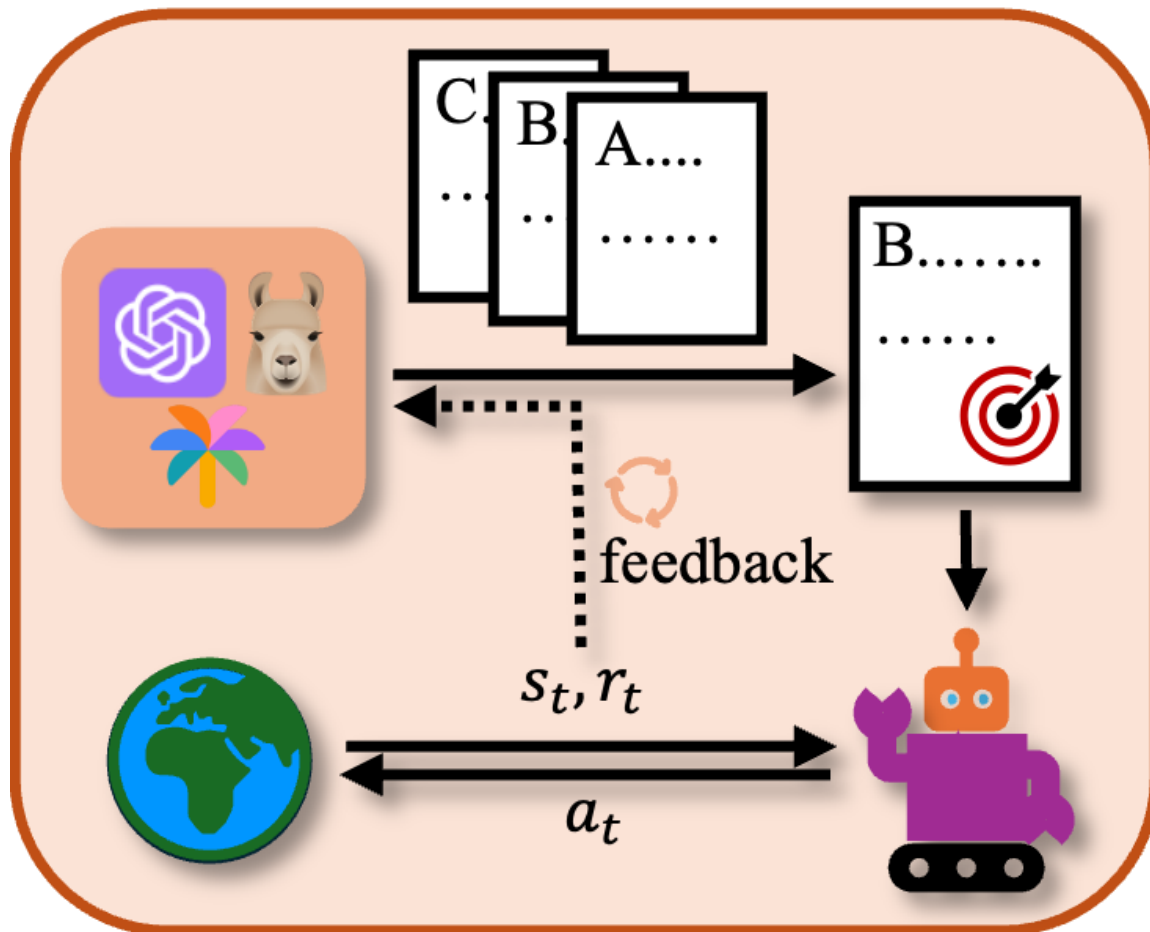
LLM/VLM as Planner: Comprehensive



- One query $\rightarrow$ full plan
- Interpreter translates each step of the plan into a goal that the agent can learn or execute
- Execution failure triggers full re-plan

2

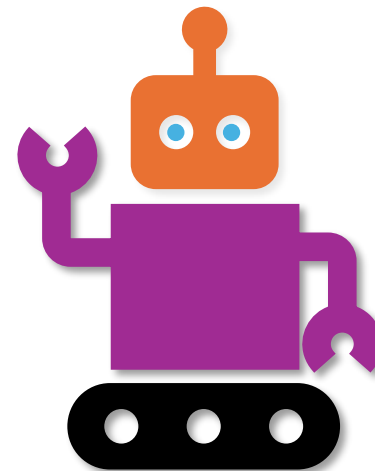
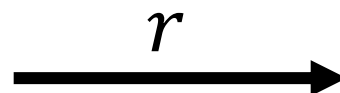
LLM/VLM as Planner: Incremental



- One query $\rightarrow$ several next-step options
- Agent executes via language-conditioned policy
- Execution feedback triggers single-step re-planning on failure

3

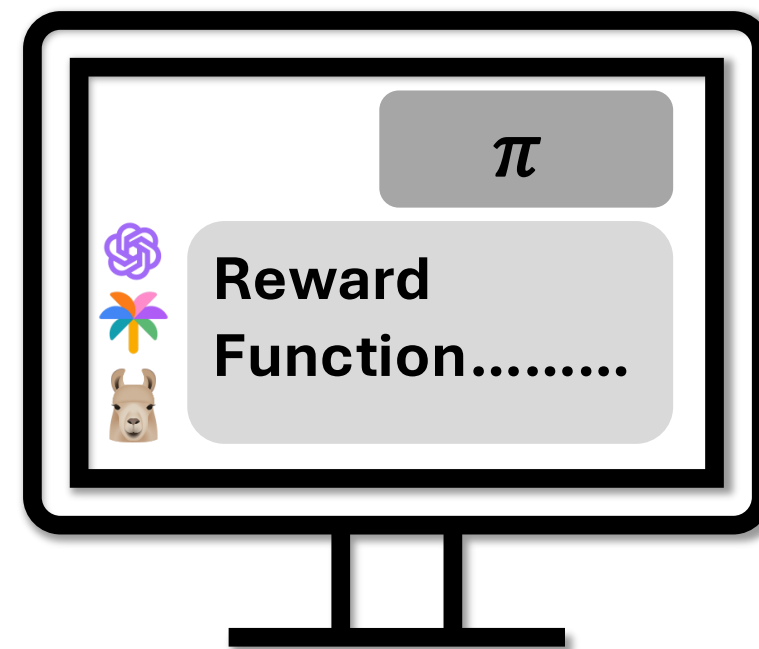
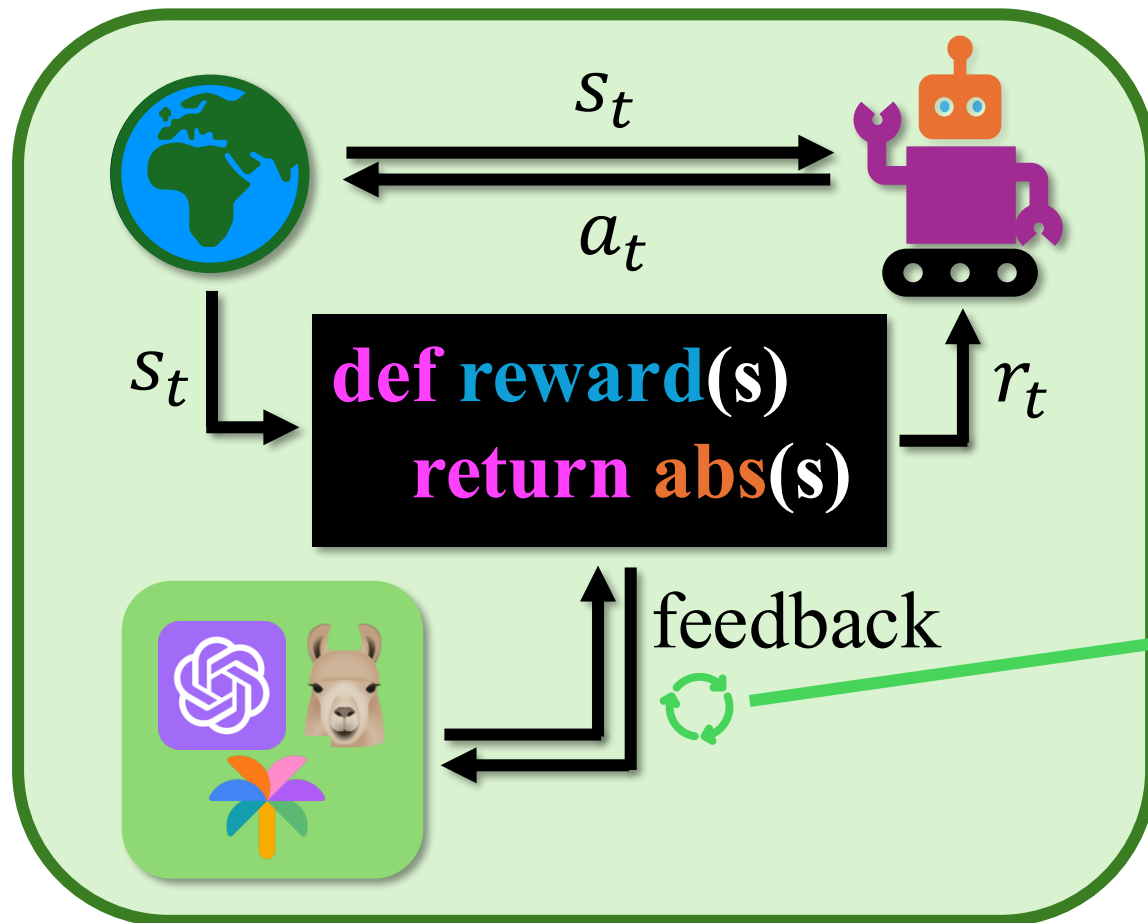
LLM/VLM as Reward



indirect / direct source of reward

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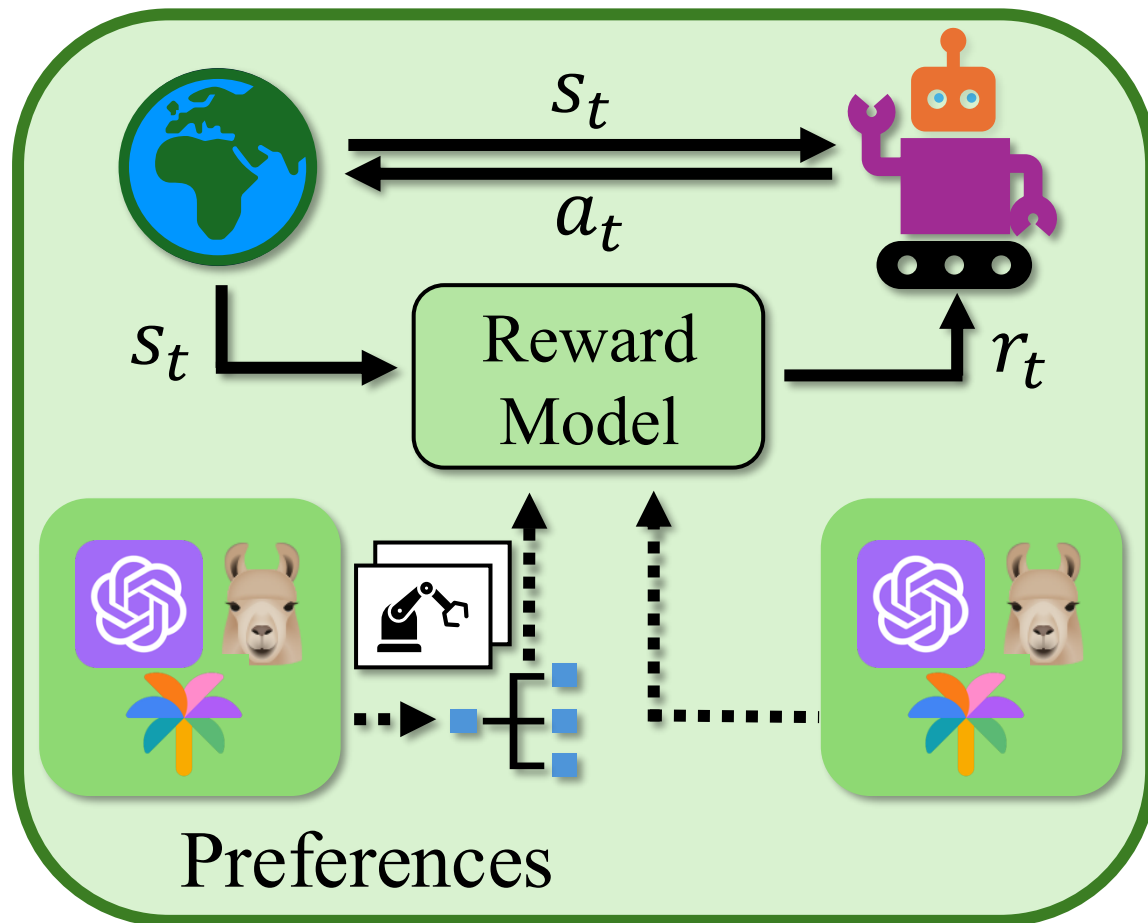
LLM/VLM as Reward: Function



In-Context Learning

3

LLM/VLM as Reward: Model



- Reward Model Learning (Left):
 - The LLM/VLM provides preferences (rankings) on trajectory sub-sequences for reward model learning
 - A learned reward model generates scalar rewards
- Reward Model (Right):
 - The LLM/VLM generates reward directly



Key Takeaways

3 LLM/VLM as Reward

